

FRAUD INTELLIGENCE PLATFORM

Behavioral Fraud Detection and Explainable Real-Time Transaction Analysis

Project Understanding Report | Team: _____ | October 2026

1. Project Overview

In one line

A web application that scores each financial transaction by comparing it with that customer's own past behavior, explains why it was scored that way, and shows which customers are linked through shared devices.

The problem we are solving

- Fraud in financial transactions is rare and hard to spot among millions of genuine payments.
- Normal behavior is different for every customer. Rs. 50,000 is routine for one person and alarming for another, so **a large transaction is not automatically fraud**.
- Real fraud usually shows up as a combination of signals: an unfamiliar device, a new or foreign location, an odd time of day, failed logins just before, many transactions in a short time.
- Some fraud involves **several accounts** using the same device. Looking at one transaction at a time misses this.
- A plain "fraud / not fraud" answer does not help an investigator. They need to know **how suspicious**, **why**, and **who else is connected**.

Our main idea

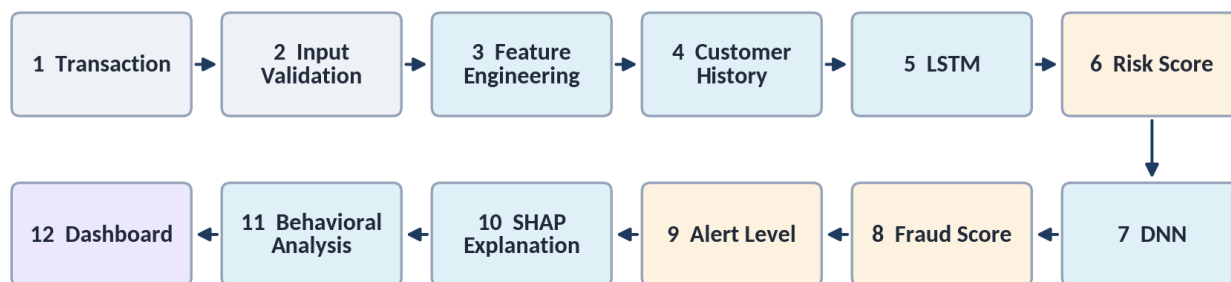
Instead of looking only at the current transaction, we compare it with the customer's previous behavior, and combine these behavioral signals with machine learning.

What our system provides

Output	What it is
Fraud Score	How suspicious this transaction looks to the model, from 0 to 100
Risk Score	How risky the customer's recent behavior was before this transaction, from 0 to 100
Alert Level	Low, Medium, High or Critical
SHAP explanation	The reasons that pushed the score up
Behavioral similarity	How closely the transaction matches the customer's normal behavior
Fraud-ring detection	Groups of customers linked by shared devices
Live transaction analysis	Score and investigate one transaction
Batch CSV scoring	Score a whole file of transactions at once
Model performance dashboard	How well the loaded model performs, and its details
PDF report	A downloadable report for a scored transaction

Technologies: Python, FastAPI, TensorFlow/Keras, SHAP and SQLite on the backend; React, Vite, Tailwind CSS and Recharts on the frontend; ReportLab for PDF reports.

2. How Our System Works



Step	What happens
1 Transaction	A transaction arrives: customer, amount, merchant category, device, location, failed logins.
2 Input validation	The backend checks that required fields are present and values are valid (for example, the amount must be above 0).
3 Feature engineering	Nine behavioral features are calculated. Each one compares this transaction with the customer's earlier transactions.
4 Customer history	The customer's past transactions are kept in time order. The 10 before this one are taken as the "recent history".
5 LSTM	A sequence model reads those 10 previous transactions.
6 Risk Score	The LSTM's output, scaled to 0 to 100: how risky the recent behavior was.
7 DNN	A second model looks at this transaction's 9 features together with the Risk Score.
8 Fraud Score	The DNN's output, scaled to 0 to 100.
9 Alert Level	Fixed bands on the Fraud Score: below 25 Low, 25 to 50 Medium, 50 to 80 High, 80 and above Critical.
10 SHAP explanation	Shows which features pushed the Fraud Score up, and by how much.
11 Behavioral analysis	Compares the transaction with the customer's usual device, city and overall pattern.
12 Dashboard	Everything is shown on screen, saved to the database, and can be downloaded as a PDF.

The two models

LSTM (the "history" model)	DNN (the "this transaction" model)
- Looks at the customer's previous 10 transactions. - Captures temporal behavioral patterns in the recent sequence (it is not claimed to predict fraud before it happens) - Produces the Risk Score.	- Takes the current transaction's 9 behavioral features. - Also receives the Risk Score from the LSTM. - Produces the Fraud Score from 0 to 100.

Fraud Score is a model score, NOT a probability

A Fraud Score of 99.9 does **not** mean a 99.9% probability of fraud. It means the model considers the transaction highly suspicious. The scores are useful for ranking and for the alert bands, not as percentages of likelihood.

3. Features and Explainability

The 9 behavioral features

Feature	Simple meaning
amount_zscore	How unusual the amount is for this customer
amount_pct_of_avg	The amount compared with the customer's average amount
hour_is_unusual	Whether the customer rarely transacts at this hour
is_new_device	Whether the customer has never used this device before
is_new_location	Whether the customer has never transacted from this location before
is_foreign_location	Whether the location is outside the list of normal home cities
category_is_unusual	Whether the merchant category is unusual for this customer
failed_logins_24h	Number of failed logins in the 24 hours before the transaction
txn_velocity_1h	Number of the customer's transactions in the last hour

Every feature is calculated only from the customer's **earlier** transactions. The DNN also receives the **Risk Score** as a tenth input.

SHAP: "Why did the model give this transaction a high score?"

SHAP is a method that splits a model's score into contributions from each feature. For every scored transaction, the dashboard lists the features that pushed the Fraud Score up, with a bar showing how much each one contributed.

Example of reasons shown for a suspicious transaction:

New device	Foreign location	Unusual transaction time	Amount far above normal
------------	------------------	--------------------------	-------------------------

This helps an investigator understand the model's decision and check each reason against the customer's record.

Important

SHAP explains the **model's decision**. It does **not** prove that the transaction is fraudulent. The reasons are a starting point for a human to verify.

Behavioral similarity

- Compares the current transaction with the customer's historical behavior, across all nine features.
- **High similarity** means the transaction looks normal for this customer.
- **Low similarity** means it looks unusual.

Deviation = 100% – Similarity

Example: similarity 0.4% means deviation 99.6%, a transaction very unlike the customer's past.

Behavioral similarity is a separate, easy-to-read indicator. It is not an input to the models. The dashboard also places the customer's usual device and home city next to the ones used in this transaction and marks each as Match or Different.

4. Dashboard Modules

The dashboard has four tabs. The header always shows which model is loaded.

1. Live Scan: investigate one transaction

- | | |
|--|--|
| <ul style="list-style-type: none">• Select a customer; their usual device, home city and history size appear• Enter the transaction (amount, category, device, location, failed logins)• The system scores it• Shows the Risk Score and the Fraud Score | <ul style="list-style-type: none">• Shows the Alert Level• Shows the SHAP reasons• Shows behavioral similarity and customer context• Generates a PDF report |
|--|--|

2. Batch Scoring: screen many transactions

- | | |
|---|---|
| <ul style="list-style-type: none">• Upload a CSV file• Every row is validated; if any row is invalid, nothing is scored and the bad rows are listed• All transactions are scored• Shows the alert distribution• Filter results by Alert Level• Download the results as CSV | <p>Required columns</p> <ul style="list-style-type: none">• customer_id• amount• merchant_category <p>Optional columns</p> <ul style="list-style-type: none">• device_id• location• failed_logins_24h |
|---|---|

Why it exists: it is useful when an analyst wants to screen many transactions instead of checking them one by one.

3. Fraud Rings: find connected customers

- Finds devices that are used by more than one customer.
- Builds customer-to-device relationships and draws them as a graph.
- Groups connected customers into clusters and shows an investigation summary for each.
- Helps investigators identify potentially coordinated activity. In our data: 29 customers in 9 clusters.

A fraud ring is a **suspicious relationship, a lead to investigate**. It is not proof of fraud: family members can share a device. This part is a rule over device IDs, not a machine-learning model.

4. Model Performance: how good is the model?

- Shows which model is currently loaded, and its version.
- Shows evaluation metrics such as recall, PR-AUC, ROC-AUC and alert rate.
- Shows model details: dataset, features, what the scores mean, and the alert bands.
- Keeps evaluation results separate from individual transaction predictions, so a metric is never mistaken for a property of one score.

5. Data, Model Evaluation and Results

Dataset

The project uses **synthetic (generated) data**, because real banking data is confidential.

Data	Used for
v1	The data behind the running application, and the training data of the production model
v2	A harder dataset, used to train and evaluate improved models
Hold-out datasets	New customers never seen in training, used to test how well a model generalizes

Why v2 was created

On v1 the model scored a perfect 1.000. We checked and found that a simple two-line rule also scored 1.000. The fraud patterns in v1 matched the features directly, so the result was unrealistically perfect. We therefore built v2 with more varied behavior:

- | | |
|--|---|
| <ul style="list-style-type: none">• Account takeover• Stolen card used online• Test charges followed by cash-out• Device takeover | <ul style="list-style-type: none">• Single high-value fraud• Fraud resembling normal activity• Ring fraud• Legitimate unusual behavior, such as travel or a new phone |
|--|---|

How we evaluated

We did not simply trust the first result. The evaluation was redesigned to remove:

- | | |
|---|---|
| <ul style="list-style-type: none">• Data leakage between the two models• Future information entering training (we now train on earlier dates and test on later ones) | <ul style="list-style-type: none">• Overlap between training and testing (final tests use completely new customers)• Threshold selection bias (thresholds are fixed before the final test) |
|---|---|

Production model = the model currently deployed. **Seed-14** = a better evaluation candidate (LSTM + DNN trained on v2), **not deployed**.

Final hold-out result

482,292 unseen synthetic transactions, containing 2,446 fraud transactions.

Measure	Production	Seed-14 (LSTM + DNN)
Recall (share of fraud caught)	27.9%	55.8%
Legitimate alerts per 1,000 transactions	10.60	9.11
Fraud transactions caught, of 2,446	683	1,365
PR-AUC	0.155	0.363
ROC-AUC	0.717	0.938

Seed-14 performed substantially better on the final synthetic hold-out, while staying within the project's alert budget of 10 legitimate alerts per 1,000. **It is still NOT the production model.**

6. Current Status, Limitations and Viva Summary

Current status	Why Seed-14 is not deployed
• Production model is the deployed default • Seed-14 is an evaluation candidate; it is not promoted automatically • Working: Live Scan, Batch Scoring, Fraud Rings, Model Performance, PDF reports • Backend tests passing (564); frontend builds successfully	• All evidence is from synthetic data • Behavior on new customers needs more work • It met the alert budget with a thin margin • More validation on realistic data is needed • Promotion needs controlled validation and rollback criteria

Limitations	Future work
• Synthetic data only • No real banking integration • Fraud Score is not a calibrated probability • New-customer (cold-start) behavior needs further validation • Fraud-ring detection is a rule-based relationship detector, not proof of fraud • An academic / research prototype	• Real transaction data • Probability calibration • Better cold-start handling • Real-time transaction streaming • Authentication • Cloud deployment • Advanced fraud graph analytics • Production monitoring

Explain our project in 60 seconds

"Our project is a Fraud Intelligence Platform that detects suspicious financial transactions by comparing them with a customer's previous behavior. The problem is that normal behavior is different for every customer, so a fixed rule does not work well. For each transaction we calculate nine behavioral features, such as a new device, a new location, an unusual time or an unusual amount. An LSTM model reads the customer's last ten transactions and gives a Risk Score. A DNN model then combines the nine features with that Risk Score and gives a Fraud Score from 0 to 100, which is a model score and not a probability. SHAP shows the reasons behind each score. A Fraud Rings module finds customers who share devices. Everything is shown in a dashboard with live scanning, batch scoring and model performance. All our data is synthetic."

Project at a glance

